LESSON 13 · CHEAT SHEET

Charts & Excel Output

EVERY CHART SCRIPT STARTS LIKE THIS

```
import matplotlib
matplotlib.use("Agg")      # no window: files
import matplotlib.pyplot as plt
import pandas as pd
from pathlib import Path
OUT = Path("my_work/output")
OUT.mkdir(parents=True, exist_ok=True)
quests = pd.read_csv("data/quests.csv")
```

THE CHART RECIPE (ALWAYS THIS ORDER)

```
plt.figure(figsize=(8, 4)) # 1 new picture
plt.bar(x, y)              # 2 draw
plt.title("...")
plt.xlabel("..."); plt.ylabel("...")
plt.xticks(rotation=30, ha="right")
plt.legend()               # if you used label=
plt.tight_layout()         # 3 no cut labels
plt.savefig(OUT / "x.png", dpi=150) # 4
```

CHART TYPES

`plt.bar(labels, values)`	vertical bars
`plt.barh(labels, values)`	horizontal bars
`plt.hist(values, bins=12)`	histogram
`plt.plot(x, y)`	line
`plt.axvline(v, linestyle="--")`	vertical marker line
`color="#1f3a5f"`, `label="Fled"`	per series

GROUPED BARS (THREE SERIES)

```
x = np.arange(len(t.index))
w = 0.27                    # 3 bars per group
plt.bar(x - w, t["Success"], w, label="Success")
plt.bar(x,     t["Fled"],    w, label="Fled")
plt.bar(x + w, t["Failed"],  w, label="Failed")
plt.xticks(x, t.index)
```

PANDAS → EXCEL

```
df.to_excel("out.xlsx", sheet_name="Data")
df.to_excel("out.xlsx", index=False)
                      # ^ no row numbers

with pd.ExcelWriter("out.xlsx") as writer:
    t1.to_excel(writer, sheet_name="By monster")
    t2.to_excel(writer, sheet_name="Top 10",
                index=False)
```

Needs `openpyxl` installed. Sheet names: max 31 chars, no `/ \ ? * [ ]`.

OPENPYXL: MAKE IT PRETTY

```
from openpyxl import load_workbook
from openpyxl.styles import Font, PatternFill
wb = load_workbook("out.xlsx")
ws = wb["Data"]                 # or wb.active
for cell in ws[1]:              # header row
    cell.font = Font(bold=True, color="FFFFFF")
    cell.fill = PatternFill("solid",
                            fgColor="1F3A5F")
for row in ws.iter_rows(min_row=2,
                        min_col=3, max_col=5):
    for cell in row:
        cell.number_format = '#,##0.00 "cr"'
ws.column_dimensions["A"].width = 16
ws.freeze_panes = "A2"
ws2 = wb.create_sheet("Chart")
# from openpyxl.drawing.image import Image
ws2.add_image(Image("chart.png"), "B2")
wb.save("out.xlsx")
```

USEFUL CELL FACTS

`ws["C2"].value`	what's in C2
`ws.max_row`, `ws.max_column`	size of the sheet
`wb.sheetnames`	list of sheet names
`"0.0%"`, `"#,##0"`, `"dd/mm/yyyy"`	other number formats

GOTCHAS

- No `plt.show()` in scripts. Save, then open the file.
- Forgot `plt.figure()`? Charts pile up on one picture.
- Labels cut off? You forgot `tight_layout()`.
- matplotlib colour `"#1F3A5F"`, openpyxl colour `"1F3A5F"`. Keyword: `color`, never `colour`.
- Workbook open in Excel = "Permission denied" when saving. Close it.